Cabinet-based Cooling Platform

Hybrid air and liquid to “future proof” your data center

DDC SOLUTIONS
a member of DAIKIN group

AI Acceleration = Infrastructure Disruption

AI Adoption

→ Exponential Growth in Compute
→ Unprecedented Rate of Next-Gen Tech Release
→ Higher Rack Densities
→ More Power Draw & Heat Output

→ Cooling Innovation Pressure
→ Infrastructure Redesign
→ Operational Risk & Cost Escalation
→ Need for Future-Proof Cooling Strategy

Future Proof Your Data Center with DDC's Cabinet-Based Cooling Platform

Rack Containment for Any Density, from H100 to Rubin

- Cool up to the highest density workloads
- Lowest cost to build
- Faster to deploy
- Less costly to run

S-Series Platform: Any Density, Any Scale, Anywhere

DDCs cabinet infrastructure supports from low density up to the **1MW+ cooling per rack**

Base

Sample equipment

H100
B100
B200
Storage Networking

GPU

(Field upgradeable)

Max

(Field upgradeable)

Sample equipment

GB200

0kW — 50 kW — 85 kW — 100kW air cooling

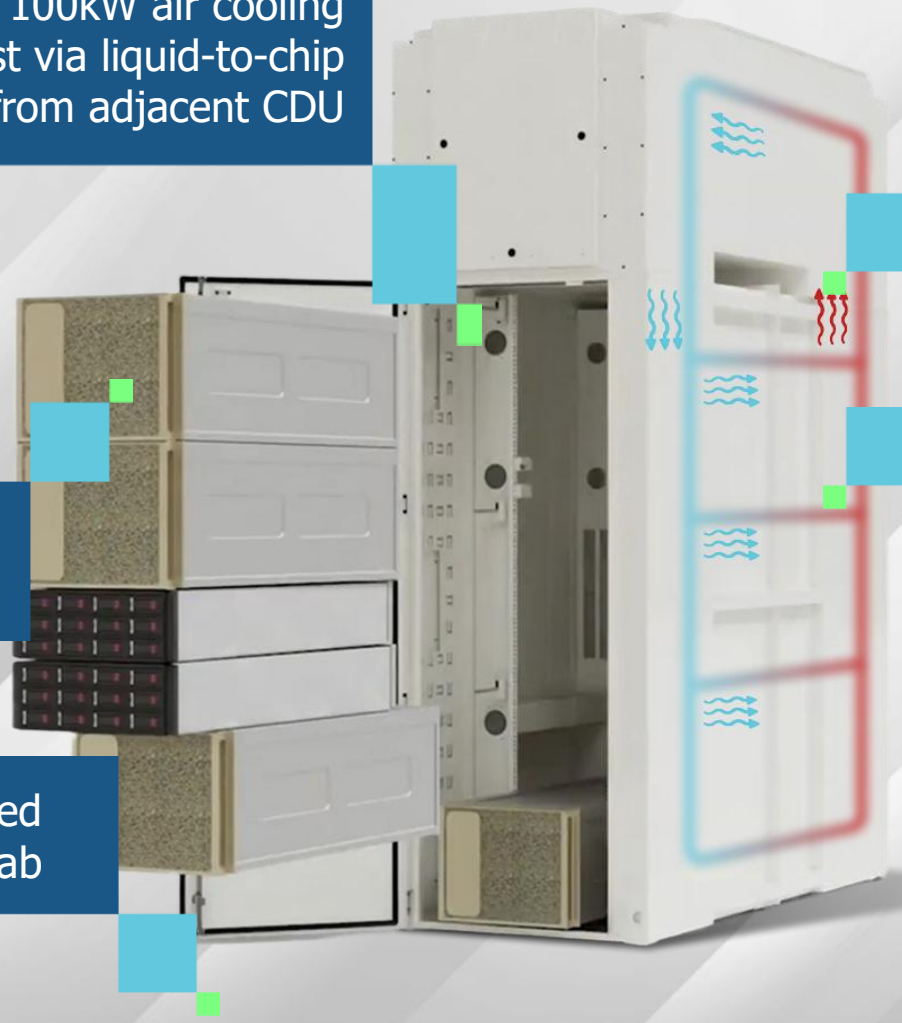
Up to 1MW+ with liquid cooling

S-Series Platform Supports

- Low, Medium or High Density Cooling
- Air Or Liquid Cooled
- Racking Servers or Roll-in Capabilities

Up to 1MW+ cooling per cabinet, with 100kW air cooling and the rest via liquid-to-chip from adjacent CDU

NEMA 3R certified for water protection



Roll-in or rack mounting capabilities

Built-in fire suppression

Works anywhere, on raised floor or slab

Air flow

Roll-in or Mounted Rack Options

- 36 inches wide cabinet width
- Supports any racking hardware
- DDC ramp fits over bottom lip of cabinet
- Swap in / out hardware as requirements change – all in the same cooling platform

Hybrid Cooling Adds CDU Cooling Capacity to Air-based Cooling

- S-Series supports Liquid-to-Chip integration for additional high-density cooling
- In this example, Chilldyne CDU 300 brings 300kW liquid-to-chip cooling, which is additive to 100kW air cooling from DDC cabinet
- **Total cooling: 400kW per rack**

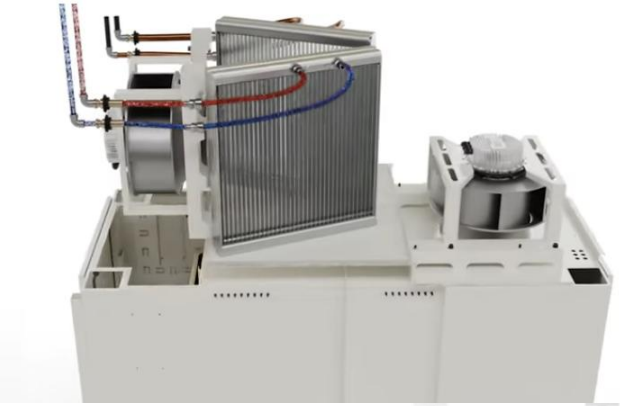
Any Density. Any Scale. Anywhere.

Supports Varying Densities from One Cabinet to the Next

Components of Cabinet Top

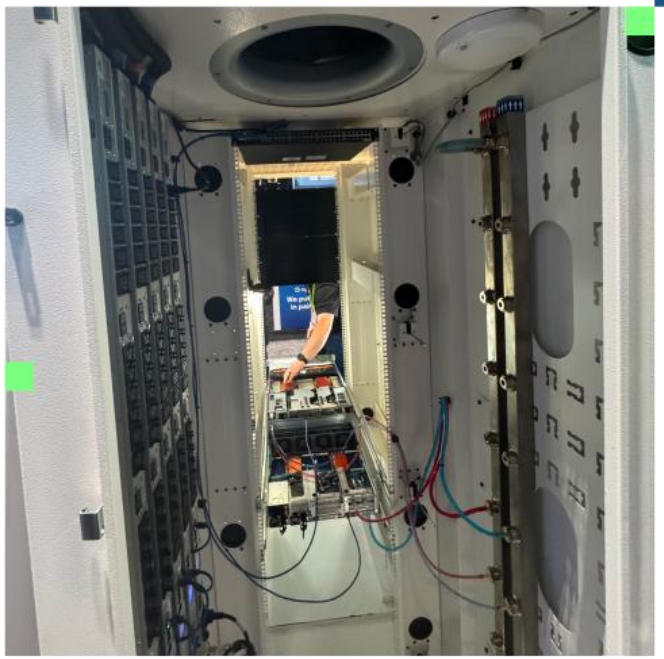
"Cabinet Top": Heat Exchanger

- Dual fans supporting circular flow and providing up to 100 kW air cooling
- Built-in fire suppression
- Flexibility in how / where heat exchange connects to
- Dual / redundant coils and fans for added resiliency
- Multiple heat exchanger options for load and PUE – all field upgradeable for larger sizes as needs evolve

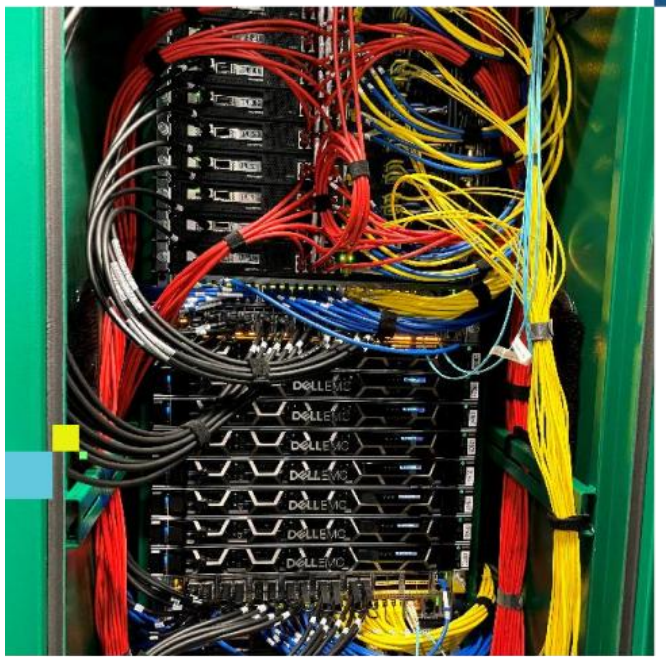


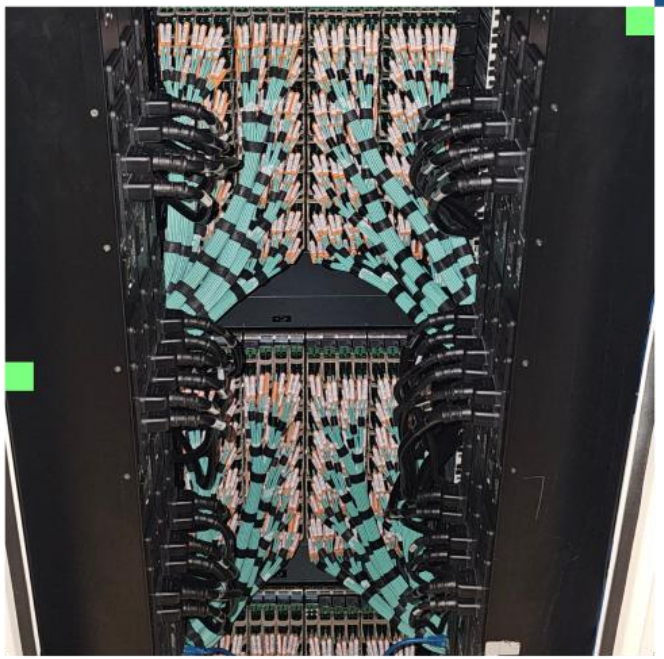
Watch the full product walk through at
ddcsolutions.com/s-series

Universal Design Supports Liquid-to-chip On Demand



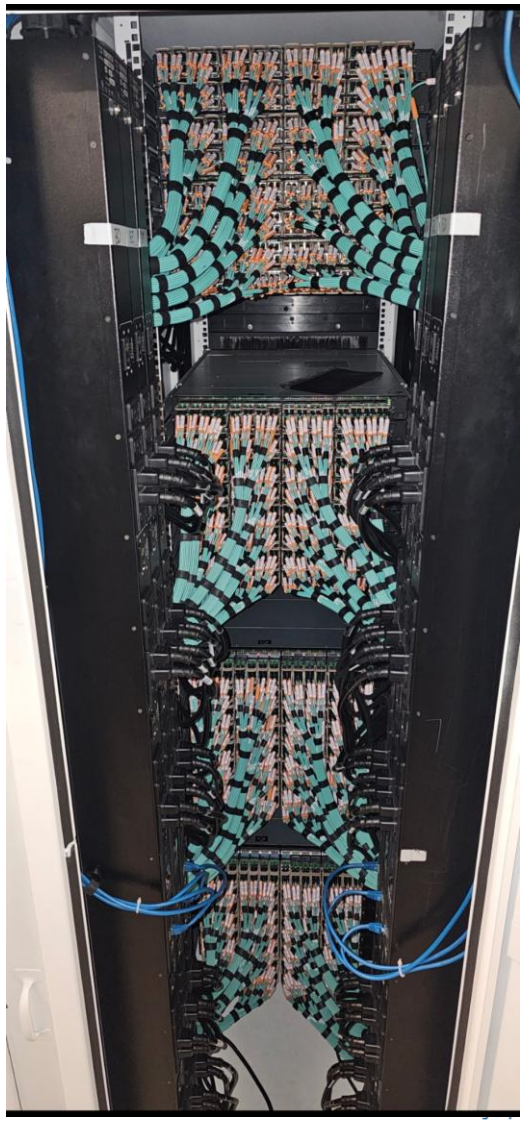
Designed for High Density Networking





- Plenty of room for cabling in rear of cabinet
- Multiple locations for power access – top, bottom, rear, front of cabinets

Recent Deployment Photos



Sample Configuration

This configuration shows an example of possible CDU placement, power access and plumbing for a data center buildout

Rapid Deployment <2 weeks (example)

In this customer example, the modular cabinet design and ease of setup allowed the customer to deploy high density cabinets in less than 2 weeks

Best in Class Asset Protection

The value of equipment in a high-density rack is routinely in the $1-2M+ range, creating a need for greater asset protection.

NEMA 3R CERTIFICATION

Ensuring the top and bottom cabinets are protected from water damage from chilled water supply and fire suppression systems (sprinklers)

BUILT-IN FIRE SUPPRESSION

Containing fire to a single cabinet and extinguishes event immediately

REDUNDANT MOTORS & COILS

Eliminating single point of failure

GRANULAR MONITORING & MANAGMENT

Controlling temperature set point, dew point, and CFM airflow to a surgical degree

Integrated DCIM Software

Dynamically monitor and manage your S-Series cabinets and other assets of your data center to reduce risk and improve efficiency with real-time, predictive environmental control from the infrastructure to the equipment level.

Data Center Level Control at the Cabinet Level

- Enhanced Real-time Asset Management
- Predictive Environmental Control
- Intuitive Dashboards and Visualization
- Improved PUE Management
- Optimal Environment Conditions
- Advanced Analytics and Fault Detection
- Power Usage
- API-Driven Communications Tools
- BMS Integration
- Dynamic Resource Allocation
- Carbon Footprint Optimization
- Customized Dashboards
- Security and Access Control

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Your Solution for Every Data Center Challenge

1. Deploy no risk air-cooled up to 100kW
2. Add Liquid-to-Chip on-demand, as needed, cabinet by cabinet to go up to 1MW+ cooling per rack
3. Adapt to changing workloads and customer profiles seamlessly
4. Improve SLA capabilities and customer performance by surgically controlling environmental conditions
5. Build cheaper, operate more efficiently from day 1
6. Upgrade existing sites with little to no impact to existing operations
7. Increase site density exponentially, unlocking untapped value per square foot
8. Deploy anywhere – raised floors, slab floors, non-data center environments

Thank You

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